

Sampling from a Target Distribution

Goal

We want to sample from a target distribution:

$$\pi(x)$$

- Often known only up to a constant:

$$\pi(x) \propto f(x)$$

- Direct sampling may be difficult

Idea

Use a simpler distribution that we can sample from.

Rejection Sampling Setup

Choose:

- Proposal distribution $g(x)$ (easy to sample)
- Constant M such that:

$$f(x) \leq Mg(x) \quad \forall x$$

Interpretation

$Mg(x)$ is an envelope over $f(x)$.

Rejection Sampling Algorithm

Repeat:

- Sample $X \sim g(x)$
- Sample $U \sim \text{Uniform}(0, 1)$
- Accept X if:

$$U \leq \frac{f(X)}{Mg(X)}$$

Otherwise

Reject and repeat.

Unconditional Acceptance Probability

Let $Y \sim g(x)$ and $U \sim \text{Uniform}(0, 1)$.

$$\begin{aligned} P\left(U \leq \frac{f(Y)}{Mg(Y)}\right) &= \mathbb{E}\left[P\left(U \leq \frac{f(Y)}{Mg(Y)} \mid Y\right)\right] \\ &= \mathbb{E}\left[\frac{f(Y)}{Mg(Y)}\right] \\ &= \int \frac{f(y)}{Mg(y)} g(y) dy \\ &= \frac{1}{M} \int f(y) dy \\ &= \frac{1}{M} \end{aligned}$$

Distribution of Accepted Samples

Let:

- $Y \sim g(x)$
- $U \sim \text{Uniform}(0, 1)$
- Accept if $U \leq \frac{f(Y)}{Mg(Y)}$

We compute the density of Y given acceptance.

Step 1: Joint probability

$$\begin{aligned}
 P(Y \in A, \text{accept}) &= \int_A g(y) P\left(U \leq \frac{f(y)}{Mg(y)} \mid Y = y\right) dy \\
 &= \int_A g(y) \frac{f(y)}{Mg(y)} dy \\
 &= \frac{1}{M} \int_A f(y) dy
 \end{aligned}$$

Conditional Density of Accepted Samples

Acceptance probability:

$$P(\text{accept}) = \frac{1}{M}$$

So:

$$\begin{aligned} P(Y \in A \mid \text{accept}) &= \frac{P(Y \in A, \text{accept})}{P(\text{accept})} \\ &= \frac{\frac{1}{M} \int_A f(y) dy}{\frac{1}{M}} \\ &= \int_A f(y) dy \end{aligned}$$

Conclusion

The accepted samples have density $f(x)$

Why Choice of M Matters

Key Result

Acceptance probability = $\frac{1}{M}$

- Number of proposals until acceptance $\sim \text{Geometric}(\frac{1}{M})$.
- Expected number of trials: $\mathbb{E}[\text{trials}] = M$.

Practical Difficulty

- Need $f(x) \leq Mg(x)$ for all x .
- g must cover entire support of f .
- Finding good (M, g) is often difficult.

Conclusion: Poor choice $\Rightarrow$ very inefficient sampling

Limitations of Rejection Sampling

- Requires $f(x) \leq Mg(x)$ for all x
- Finding good $g(x)$ and small M is difficult
- Efficiency drops in high dimensions

Key Issue

Many proposals are rejected $\Rightarrow$ wasteful

Idea

Instead of independent samples, construct a **Markov chain** whose stationary distribution is $\pi(x)$.

Class Problems: Rejection Sampling

- **Problem 1: Uniform Proposal** Find the value of M .

Target: $f(x) = 2x, x \in [0, 1]$

Proposal: $g(x) = 1, x \in [0, 1]$

- **Problem 2: Acceptance Rule**

Target: $f(x) = 2x, x \in [0, 1]$

Proposal: $g(x) = 1, x \in [0, 1]$

- **Problem 3: Thin tails**

Target: $f(x) = e^{-|x|}, x \geq 0$

Proposal: $g(x) = \mathcal{N}(0, 1)$

- **Problem 4: Support Mismatch**

Target: $f(x) = \mathcal{N}(0, 1)$

Proposal: $g(x) = \text{Uniform}[-1, 1]$

- **Problem 5: High-Dimensional Case**

Target: $f(x) = \mathcal{N}(0, I_d)$

Proposal: $g(x) = \text{i.i.d Exponential}(\theta)$

Markov Chain Monte Carlo (MCMC)

Idea

Generate samples using a Markov chain:

$$X_0 \rightarrow X_1 \rightarrow X_2 \rightarrow \dots$$

- Each X_{t+1} depends only on X_t
- After many steps, distribution converges to $\pi(x)$

Goal

Design transitions so that:

$$\pi(x) \text{ is stationary}$$

Detailed Balance Condition (Reversibility)

A sufficient condition for $\pi(x)$ to be stationary:

$$\pi(x)P(x \rightarrow y) = \pi(y)P(y \rightarrow x)$$

Interpretation

Flow from $x \rightarrow y$ equals flow from $y \rightarrow x$

Strategy

Construct transition probabilities satisfying this condition

Continuous-State Markov Chains (Brief Overview)

- State space is continuous: $X_t \in \mathbb{R}^d$
- Cannot use transition *matrix* anymore
- Use a **transition density** / **kernel**:

$$P(x \rightarrow y) = K(y \mid x)$$

- Evolution:

$$X_{t+1} \sim K(\cdot \mid X_t)$$

Key Idea

Instead of probabilities of jumps, we now work with **densities**

Stationary Distribution (Continuous Case)

A distribution $\pi(x)$ is stationary if:

$$\pi(y) = \int \pi(x) K(y | x) dx$$

- Continuous analogue of $\pi P = \pi$
- Distribution remains unchanged after one step

Interpretation

If $X_t \sim \pi$, then $X_{t+1} \sim \pi$

How to Ensure Stationarity?

A sufficient condition: **Detailed Balance**

$$\pi(x) K(y | x) = \pi(y) K(x | y)$$

- Stronger than stationarity
- Much easier to enforce in practice

Why this matters

Metropolis–Hastings is designed to satisfy this condition

Why Metropolis–Hastings?

- Rejection sampling can be inefficient
- Difficult to find good envelope $Mg(x)$

Idea

Construct a **Markov chain** whose stationary distribution is $\pi(x)$

- Reuse previous samples
- Avoid complete rejection of proposals
- Avoid finding an envelope

Markov Chain Construction

Generate a sequence:

$$X_0, X_1, X_2, \dots$$

- X_{t+1} depends only on X_t
- Transition kernel: $P(x \rightarrow y)$

Goal

Design transitions such that:

$$\pi(x) \text{ is stationary}$$

Proposal Distribution

At state $X_t = x$:

- Propose a candidate:

$$Y \sim q(y \mid x)$$

- $q(y \mid x)$ is easy to sample from.

Examples

- Gaussian random walk
- Independence proposal

Acceptance Probability

Accept proposed move with probability:

$$\alpha(x, y) = \min \left(1, \frac{\pi(y) q(x | y)}{\pi(x) q(y | x)} \right)$$

Using unnormalized density

If $\pi(x) \propto f(x)$:

$$\alpha(x, y) = \min \left(1, \frac{f(y) q(x | y)}{f(x) q(y | x)} \right)$$

Key: Normalizing constant cancels

Metropolis–Hastings Algorithm

Given $X_t = x$:

- ① Sample $Y \sim q(y \mid x)$
- ② Sample $U \sim \text{Uniform}(0, 1)$
- ③ If

$$U \leq \alpha(x, y)$$

set $X_{t+1} = Y$

- ④ Otherwise, set $X_{t+1} = x$

Transition Kernel

The transition probability is:

$$P(x \rightarrow y) = \begin{cases} q(y | x) \alpha(x, y), & y \neq x \\ 1 - \int q(z | x) \alpha(x, z) dz, & y = x \end{cases}$$

Interpretation

- Move to y with acceptance probability
- Otherwise stay at x

Detailed Balance: Case 1

Case 1:

$$\frac{\pi(y) q(x | y)}{\pi(x) q(y | x)} \leq 1$$

Then:

$$\alpha(x, y) = \frac{\pi(y) q(x | y)}{\pi(x) q(y | x)}$$

So:

$$\begin{aligned} \pi(x) P(x \rightarrow y) &= \pi(x) q(y | x) \alpha(x, y) \\ &= \pi(x) q(y | x) \frac{\pi(y) q(x | y)}{\pi(x) q(y | x)} \\ &= \pi(y) q(x | y) \end{aligned}$$

Detailed Balance: Case 2

Case 2:

$$\frac{\pi(y) q(x | y)}{\pi(x) q(y | x)} > 1$$

Then:

$$\alpha(x, y) = 1$$

So:

$$\pi(x)P(x \rightarrow y) = \pi(x) q(y | x)$$

For reverse move:

$$\alpha(y, x) = \frac{\pi(x) q(y | x)}{\pi(y) q(x | y)}$$

Hence:

$$\begin{aligned} \pi(y)P(y \rightarrow x) &= \pi(y) q(x | y) \alpha(y, x) \\ &= \pi(y) q(x | y) \frac{\pi(x) q(y | x)}{\pi(y) q(x | y)} \\ &= \pi(x) q(y | x) \end{aligned}$$

Detailed Balance

Metropolis–Hastings satisfies:

$$\pi(x)P(x \rightarrow y) = \pi(y)P(y \rightarrow x)$$

Why important?

Ensures $\pi(x)$ is a stationary distribution

Key Idea

Acceptance probability corrects asymmetry in q

Intuition

- If $\pi(y) > \pi(x)$:
 - Move is always accepted
- If $\pi(y) < \pi(x)$:
 - Move accepted with probability:

$$\frac{\pi(y)}{\pi(x)} \times \frac{q(x | y)}{q(y | x)}$$

Insight

Allows exploration but favors high-density regions

Properties of MH

- Produces correlated samples
- Requires burn-in period
- Thinning (keeping every k -th sample) can reduce correlation
- Converges under mild conditions:
 - irreducibility
 - aperiodicity

Advantage

Works even when $\pi(x)$ is known only up to a constant

Gaussian Random Walk Proposal

A common choice:

$$q(y \mid x) = \mathcal{N}(y; x, \sigma^2 I)$$

- Propose:

$$Y = x + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2 I)$$

- Small random perturbation around current state

Interpretation

Local exploration of the state space

Symmetry of Gaussian Proposal

For Gaussian random walk:

$$q(y | x) = q(x | y)$$

Reason

$$\mathcal{N}(y; x, \sigma^2 I) = \mathcal{N}(x; y, \sigma^2 I)$$

Consequence

Acceptance probability simplifies:

$$\alpha(x, y) = \min \left(1, \frac{\pi(y)}{\pi(x)} \right)$$

Behavior of Gaussian Random Walk

- If σ is small:
 - High acceptance rate.
 - Slow exploration
- If σ is large:
 - Low acceptance rate.
 - Many rejections

Trade-off

Choose σ to balance:

- exploration vs acceptance

Example: Using Unnormalized Density

Suppose:

$$\pi(x) \propto f(x)$$

Then acceptance probability:

$$\alpha(x, y) = \min \left(1, \frac{f(y)}{f(x)} \right)$$

Key Advantage

No need to compute normalizing constant

Independence Sampler

Proposal does not depend on current state:

$$q(y \mid x) = q(y)$$

Acceptance probability:

$$\alpha(x, y) = \min \left(1, \frac{\pi(y) q(x)}{\pi(x) q(y)} \right)$$

Interpretation

Global proposals instead of local moves

Behavior of Independence Sampler

- Works well if $q(y) \approx \pi(y)$
- Poor choice $\Rightarrow$ many rejections

Comparison

- Gaussian random walk: local exploration
- Independence sampler: global exploration

Class Problems: Rejection Sampling (Advanced)

- **Problem 1: Optimal Proposal within a Family**

Target: $f(x) = e^{-x^2/2}$

Proposal: $g(x) = \text{Laplace}(0, b)$

Find $M(b)$ and optimize over b

- **Problem 2: Mixture Proposal Design**

Target: $0.5\mathcal{N}(-3, 1) + 0.5\mathcal{N}(3, 1)$

Proposal: $\mathcal{N}(0, \sigma^2)$

Find where the optimal value of M for this proposal.

Improve using mixture proposal

- **Problem 3: Adaptive Envelope**

Estimate M using observed $\max \frac{f(x)}{g(x)}$

What issues arise?

Class Problem: Transformation and Proposal Design

Problem 4: $f(x) = xe^{-x}$, $x > 0$

- **(A)** Let $g(x) = e^{-x}$. Compute $\frac{f(x)}{g(x)}$ and show that no finite M exists.
- **(B)** Let $y = \log x$. Show that:

$$f_Y(y) = e^{2y} e^{-e^y}$$

- **(C)** Consider

$$g_Y(y) = \frac{\lambda}{2} e^{-\lambda|x|}$$

- **(D)** Compute $\frac{f_Y(y)}{g_Y(y)}$ and show that a finite M exists for $\lambda \geq 2$.

Class Problem: Non-Reversible Chains

Problem 5: Non-Reversible Markov Chain

Consider a Markov chain on state space $\{1, 2, 3, 4\}$ with transitions:

- $1 \rightarrow 2$ with probability 1
 - $2 \rightarrow 3$ with probability 1
 - $3 \rightarrow 1$ with probability $\frac{1}{2}$, and $3 \rightarrow 4$ with probability $\frac{1}{2}$
 - $4 \rightarrow 2$ with probability 1
-
- **(A)** Show that the chain is irreducible
 - **(B)** Show that the chain is aperiodic
 - **(c)** Find the stationary distribution π
 - **(D)** Verify that detailed balance does *not* hold

Class Problems: Metropolis–Hastings

- **Problem 6: MH vs Rejection Sampling**

Independence proposal: $q(y|x) = g(y)$

When does MH behave like rejection sampling?

- **Problem 7: Proposal Scale Effects**

$Y = x + \epsilon, \epsilon \sim \mathcal{N}(0, \sigma^2)$

Analyze behavior as $\sigma \rightarrow 0$ and $\sigma \rightarrow \infty$

- **Problem 8: Sticky Chains**

Construct proposal with high self-transition probability

Effect on convergence?

- **Problem 9: Acceptance Geometry**

Interpret $\alpha(x, y) = \min(1, r)$

Relation to optimization vs sampling